Assessing Ranking Fidelity of Competition Structures via Weighted Mutual Information

Abstract

We evaluate multiple competitions structures to assess their effectiveness in accurately reproducing the true underlying ranking of participating teams. Using repeated simulations for each structure, we compare teams' true performance ranks to their simulated outcomes. To quantify the accuracy of each format, we calculate weighted mutual information between true and simulated rankings across a large number of replicates. "Salient" weights are applied to look at only the top i out of n total teams/competitors in a given competition structure. This information-theoretic approach allows us to objectively compare formats and identify which structures most effectively preserve rank order. Results are visualized through comparative plots, providing clear insights into the trade-offs and strengths of each competition design.

| Keywords: tournament structures, mutual information

1 Introduction

There are hundreds of different competitions structures that dictate how a set of teams compete against each other to determine an overall ranking. Common structures include bracket tournaments where if a team wins their competition they advance to the next round. Bracket tournaments can be run as single elimination where competitors advance or are eliminated based on a single match (e.g. NCAA College Basketball Tournament, National Football League (NFL) playoffs, major tennis tournaments) or where each round is contested as a best of k series (e.g. $k = 7$: National Basketball Association (NBA) playoffs, National Hockey League (NHL) playoffs, k varies by round: Major League Baseball (MLB), Women's National Basketball Association (WNBA)) and the series winner advance while the series loser is eliminated. Additional styles of the bracket tournaments include variations where competitors are not eliminated after a single loss such as double elimination tournaments where competitors continue until they have lost twice. Within double elimination tournament, there are several variations. For instance, the NCAA College Wrestling tournament is a double elimination tournament, however, if a competitor loses a match they fall into the consolation bracket and the best they can finish after a loss before the finals is 3rd place making this not

a "true" double elimination format. Olympic judo uses a variant of this system with single elimination until the quarterfinals followed by a consolation bracket consisting of the four quarterfinal losers who still have a chance for a bronze medal[1].

"True" double elimination - a bracket tournament where a competitor can still finish ranked first as long as a competitor has not lost two games - is used in NCAA baseball and softball for certain rounds in their postseason. Additional variants of double elimination include double elimination brackets with a "true third" match which includes an additional match contested between the competitors ranked 2nd (loser of the finals) and 3rd (winner of the consolation bracket) at the end of the tournament *if they have not already competed against each other*. There is also a variation of double elimination where only those competitors who have lost to a competitor that makes the finals are placed into the consolation bracket and given the chance to compete for third place. This style of competition is used in Olympic wrestling (both freestyle and Greco-Roman)[2].

In bracket style competitions, only a handful of the match-ups that are possible actually occur. When every possible competitive pairing occurs, this is referred to as a round-robin structure where every competitor plays every other competitor the same number of times. Single round-robin, where was possible match-up occurs exactly once, is used in, for examples, the group stage of the FIFA World Cup to create a ranking to determine which teams advance to the next round. Double round-robin, where each possible pairing occurs twice, is used in association football in English Premier League (EPL) and in chess during the FIDE candidates tournament (a tournament that decides who will challenge the world champion). For the sake of brevity, we review only a small number of possible competition structures, but the interested reader can find a more complete list of competition structures in Devriesere and Goossens (2025), Appleton (1995) and Csato (2021).

A natural question of interest after reviewing different types of competition structures is: which competition structure is best? As with most complicated questions, the answer is "it depends". One commonsense conceptualization for the best competition structure is that the final rankings produced by a competition structure should closely match the true rankings of the strength of the competitors. That is, a good competition structure should rank the true best team first, the true second best team second, and so on, until finally ranking the true worst team last. We refer to this property of a competition structure as the efficacy of the structure (Lasek and Gagolewski (2018), Sziklai, Biró, and Csató (2022)). If one were only interested in correctly ranking the true best competitor as first, with the ranks of the remaining competitors viewed as inconsequential, we refer to this here are effectivity

[1] in Olympic judo they award two bronze medals to the two competitors who each win their side of the consolation bracket

[2] As in Olympic judo, Olympic wrestling also award two bronze medals to the winners of each side of the consolation bracket

(Glenn (1960)). While there are many competition structures where efficacy and effectivity are useful measures[3], there are many other competitions where the correct ranking of more than one competitor, but not all the competitors is of import. For instance, in Olympic competition, the goal of a competition structure is to correctly rank the top 3 competitors to award gold, silver, and bronze medals to[4]. In many American sports, for instance the NFL, the goal of the competition structure of the regular season is to rank the top 7 teams in each conference to choose the playoff teams with the ranking of the teams 8 through 16 in each conference inconsequential in determining a Super Bowl champion[5]. We coin the term "salient ranks" to refer to the ranks of a competition structure that matter. That is the ranks 1, 2, and 3 in Olympic competition are the salient ranks in that competition structure. In this notation, efficacy is a measure with with salient ranks 1 through n where n is the number of competitors, and effectivity is a measure where 1 is the only salient rank.

If the only goal of a competition structure was high efficacy of the salient ranks, the best way to achieve this would be to play each pair of possible match ups of competitors as many times as possible. Of course, this is impossible due to many factors including time constraints, cost constraints, and wear and tear on competitors in high impact sports (e.g. boxing, American football, etc.). In addition, competitions where the true best team is guaranteed or nearly guaranteed to win may be less appealing to fans and may drive down demand as discussed in Johnson and Fort (2022).

In light of this, many different competition structures have been proposed, several of which are mentioned previously. Given these many competition structures, ideally there would be a way comparing the performance of different competition structures to each other.

Here, we propose a novel metric for evaluating different competition structures using an information theoretic approach. As an analogy, we view the results of a tournament as a transmitted message, and we use information theory to assess how accurately the message (i.e. the tournament results) matches the true transmission (i.e. the true ranking of competitors).

Specifically, we use weighted mutual information with a weights based on squared error loss[6] between the ranks generated by a competition structure and the true ranks of the competitors. Additionally, we include an indicator vector to specify the salient ranks, allowing us to compare different competition structures with different choices for salient ranks.

[3]efficacy is a useful measure for the EPL, for example, as the ranking of the teams matters at both the high end (i.e. entry into special tournaments) and the low end (i.e. relegation status) and effectivity is a useful measure for the NCAA College Basketball tournament (i.e. main goal is to crown a champion)

[4]Some sports award two bronze medals (e.g. wrestling, judo, boxing, tae kwon do)

[5]The ranking of teams outside of the playoffs does matter for draft purposes though

[6]other loss functions can easily be incorporated into our method

The remainder of this manuscript presents the mathematical foundation of the proposed measure followed by a simulation study estimating out proposed measure for different types of competition structures with varying choices of salient ranks. We close by offering a summary of the work and ideas for future directions.

2 Methods

Consider a set of n teams indexed from $i = 1, 2, \cdots, n$ each with an associated true strength parameter $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_n)$ such that $(\theta_1 > \theta_2 > \ldots > \theta_n)$. Next, let $r(\boldsymbol{\theta})$ be the indexes of $\boldsymbol{\theta}$ when θ is sorted from largest to smallest so that $r(\boldsymbol{\theta}) = \{1, 2, \ldots, n-1, n\}$. Next define $T(\phi, s)$ as a tournament with schedule ϕ and seeding structure s. ϕ contains all the information about the scheduling of teams, which could be fully known prior to the tournament (*e.g.* round robin) or determined as the tournament progresses (*e.g.* single elimination tournament). We then let $\hat{r}(T(\phi, s), \boldsymbol{\theta})$ be an n-dimensional vector-valued random variable that gives the results of a tournament as a vector of the indexes of the vector $\boldsymbol{\theta}$. The index in the first position of the vector $\hat{r}(T(\phi, s), \boldsymbol{\theta})$ indicates the index of the team that finished first, the index in the second position of the vector $\hat{r}(T(\phi, s), \boldsymbol{\theta})$ indicates the index of the team that finished second, and so on.

For example, in a $n = 4$ team tournament, $\hat{r}(T(\phi, s), \boldsymbol{\theta}) = (3,\ 2,\ 1,\ 4)$ indicates that the team with index 3 (i.e. the true third best team) won the tournament, the true second best team finished second, the true best team finished third and the true 4th team finished 4th in that particular tournament. If $\hat{r}(T(\phi, s), \boldsymbol{\theta}) = (1,\ 2,\ 3,\ 4)$, this means that the random outcome of the tournament was the same as the true ordering of the teams.

If one views the true ranking of the teams in a tournament as a message to be sent to a receiver and the outcome of the tournament as a message that is received, we can measure the "goodness" of a tournament in terms of its ability to accurately transmit the true ranking. We can then use concepts from information theory to assess the ability of a tournament to correctly rank teams. Specifically, we start with the concept of mutual information (Guiasu (1977)). For two random variables X and Y mutual information is defined as:

$$I(X; Y) = \sum_{y \in Y} \sum_{x \in X} p(x, y) \log \frac{p(x, y)}{p(x)\, p(y)}$$

In our setting, we replace X and Y with $r(\boldsymbol{\theta})$ and $\hat{r}(T(\phi, s), \boldsymbol{\theta})$, and we could seek to estimate $I(r(\boldsymbol{\theta}), \hat{r}(T(\phi, s), \boldsymbol{\theta}))$. For simplicity, we drop the arguments from the functions r and $\hat{r}$ for ease of exposition.

We note here that r is not a random variable, however, since the indexing of the teams in

the vector $\boldsymbol{\theta}$ is arbitrary (we use 1, 2, 3, etc. for convenience, but any indexing is valid), we can view this a random variable where all permutation so of the n teams are equally likely. Therefore when calculating the mutual information in this setting, we set $p(r) = \frac{1}{n!}$. By a similar argument we set $p(\hat{r}) = \frac{1}{n!}$.

We define the mutual information of r and $\hat{r}$ to be:

$$I(r, \hat{r}) = \sum_{r} \sum_{\hat{r}} p(r, \hat{r}) \log \frac{p(r, \hat{r})}{p(r)p(\hat{r})} = n! \sum_{\hat{r}} p(r, \hat{r}) \log \frac{p(r, \hat{r})}{(\frac{1}{n!})^2}$$

While there are $n!$ different permutations for the result of r, we don't need to sum across these as any specific choice of r is just an arbitrary labeling of the true strength parameters vector $\boldsymbol{\theta}$. So given r, the distribution of $\hat{r}$ is the same up to the labeling. Therefore, we only need to consider a single permutation of r, we compute the the quantity inside the summation, and then multiple by $n!$ (effectively summing across r).

In order to compute this quantity, we need to compute $p(r, \hat{r})$. This probability is found by calculating the probability of a given permutation of $\hat{r}$ and r is assumed to be the permutation from $\{1, 2, \ldots, n\}$. We estimate $p(r, \hat{r})$ empirically through simulation.

However, using mutual information in this form does not suit our needs in this setting. The problem is that mutual information in this form will yield high values of mutual information when the output from the tournament $\hat{r}$ is highly consistent *even if the ranking from the tournament is incorrect*. As an example, if $n = 4$ and the true order of $\boldsymbol{\theta}$ is $r = \{1, 2, 3, 4\}$ and $p(\hat{r} = \{4, 3, 2, 1\}) = 1$ will be the same mutual information as when $r = \{1, 2, 3, 4\}$ and $p(\hat{r} = \{1, 2, 3, 4\}) = 1$ and in both of these cases the mutual information will be maximized at:

$$I(r, \hat{r}) = n! \sum_{\hat{r}} p(r, \hat{r}) \log \frac{p(r, \hat{r})}{\left(\frac{1}{n!}\right)^2} = n! log((n!)^2)$$

and for the specific case when $n = 4$ would be:

$$4! log_2(4!^2) = 24 * log_2(24^2) = 220.0782$$

In order to alleviate this problem, we instead consider *weighted* mutual information (Guiasu (1977)). We want to give more weight to permutations from $\hat{r}$ that are "closer" to r.

In general, any loss function $l(r, \hat{r})$ can be used to define the weigting function:

$$w(r,\hat{r}) = \begin{cases} \frac{1}{l(r,\hat{r})}, & r \neq \hat{r} \\ 1, & r = \hat{r} \end{cases}$$

Here though, we choose to use squared error loss:

$$w(r,\hat{r}) = \begin{cases} \frac{1}{\hat{r}'r}, & r \neq \hat{r} \\ 1, & r = \hat{r} \end{cases}$$

However, because the number of teams that need to be accurately ordered differs based on the choice of salient ranks, a second set of weights is defined, m, is used to indicate which ranks are salient. We call these the "salient weights" to differentiate from the other weighting function. As an example, for a four-team tournament, if the organizer decides that only the winner is important, then $m = \{1, 0, 0, 0\}$ (i.e. essentially measuring effectivity). While here we only consider indicators (i.e. 0's or 1's) in the salient weights vector, there is no reason why you couldn't weight the ranks by relative levels of importance. For instance, letting $m = \{3, 1, 0, 0\}$ indicates that the salient ranks are 1 and 2, but getting the true best competitor as the winner is more important than ranking the true second best competitor second.

With the addition of the weighting functions, the formula for mutual information becomes:

$$wI_m(r,\hat{r}) = \sum_r \sum_{\hat{r}} w(r,\hat{r}) * m * p(r,\hat{r}) \log \frac{p(r,\hat{r})}{p(r)p(\hat{r})}$$

Because this mutual information is unitless, the formula can be standardized to be between 0 and 1 as follows:

$$\frac{wI_m(r,\hat{r})}{\max\{H(r), H(\hat{r})\}}$$

where $H(.)$ is the usual Shannon entropy. Competition structures with higher levels of efficacy are indicated when our measure is closer to 1. Likewise, competition structures that are near 0 by our proposed metric indicate low levels of efficacy.

3 Results

While we have explored many more competition scenarios, for the sake of space we present only results of tournaments with $n = 8$ with the distribution of true strengths distributed as equally spaced quantiles of a standard normal distribution unless otherwise noted. In order to simulate each game we compute the probability that team j will beat team k by taking

the inverse logit transformation of the differences in their strength parameters such that

$$p_{jk} = \frac{e^{\theta_j - \theta_k}}{1 + e^{\theta_j - \theta_k}}$$

where p_{jk} is the probability that competitor j deats competitor k and θ_j and θ_k are the strength parameters of teams j and k, respectively.

We focus on several simple competition structures focused mainly on variations of bracket tournaments to demonstrate the proof of concept of our proposed metric. In all cases, where seeding is needed (i.e. all the bracket structures), the seeds are considered to be known and correct reflecting each team's true rank. Note that in bracket structures, the final rankings of teams not reaching the finals was randomly assigned among possible ranks. For example, if a team in a single elimination bracket lost in the semi-finals, their possible ranks are either 3 or 4 and are equally likely; if a team loses in the quarterfinals in that same tournament , their final rank is randomly assigned from the set of ranks 5 through 8. Each competition structure was simulated 10,000 times and those results were used to estimate $wI_m(r, \hat{r})$.

Specifically, we examined the following structures:

- Single elimination bracket, usual seeding (1 vs 8, 4 vs 5, 3 vs 6, 2 vs 7)
- Single elimination bracket, bad seeding (1 vs 2, 3 vs 4, 5 vs 6, 7 vs 8)
- Double elimination bracket, usual seeding (1 vs 8, 4 vs 5, 3 vs 6, 2 vs 7)
- Single elimination bracket, usual seeding (1 vs 8, 4 vs 5, 3 vs 6, 2 vs 7) matches determined by a coin toss.

- Round Robin, normally distributed strengths
- Every Possible Permutation: All possible 8! = 40320 permutations of rankings are equally likely.

Figure 1 shows the results of the estimated values of $wI_m(r, \hat{r})$ on the y-axis (note the y-axis is on a log_{10} scale) while the x-axis represents the number of salient weights that were considered. Intuitively, of the competition structures considered here, the round robin (shown in orange) should be the best as it consists of the most match-ups by far of any structure considered here. Note that it dominates all of the bracket structures regardless of the choice of the number of salient ranks.

The single and double elimination bracket structures with traditional seeding (red and light blue, respectively) are nearly identical and both drastically better than the single elimination tournament with a bad seeding structure (shown in black). The green curve preserves the structure of a single elimination bracket and has the correct seeding, but picks the winner based on a coin flip rather than weighting simulated outcomes based on the strength of the competitors, whereas the yellow curves shows results where every possible permutation

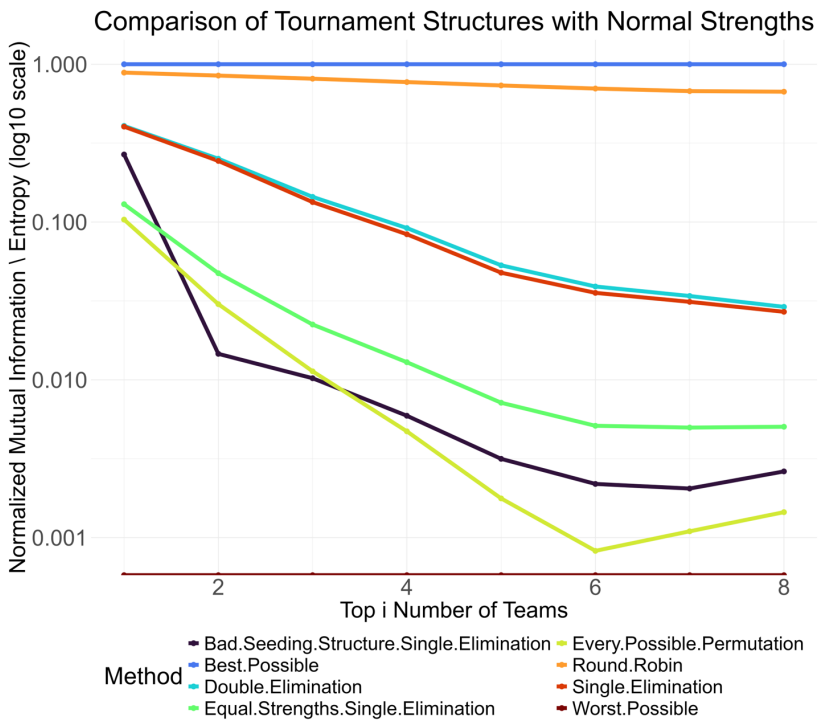

Figure 1: This figure compares the normalized weight mutual information (on log10 scale) of various competition structures against the number of salient ranks.

of rankings are equally likely. The green curve dominates the yellow curve here because there are certain final rank orderings that are not possible in bracket determined by coin flips because of the structure. So the information that is generated from the tournament is based solely on the seeding structure of the tournament. Whereas the yellow curve is based on ranks generated from a tournament with essentially no structure at all that happens to generate reasonable rankings occasionally.

Probably most notable in this figure is the black curve briefly dipping below the yellow and green curves. What is happening here is that the single elimination tournament with bad seeding (in black) is better when the there is only a single salient rank. However, when the number of salient ranks is 2, this drops *below* the two curves that are generates based on coin flip outcomes and completely random outcomes. This is because in this structure the 1 and 2 seed both play each other in the first round meaning that one of them will be eliminated immediately. Thus under this seeding mechanism, the competition structure simply cannot generate a correct result when the number of salient ranks is 2. This structure is still below both green and yellow curves when the number of salient ranks is 3, then recovers and climbs above those curves when the number of salient ranks is 4 and continues for the values of salient ranks 5 through 8.

Figure 2 shows the same red and light blue curves representing single and double elimination,

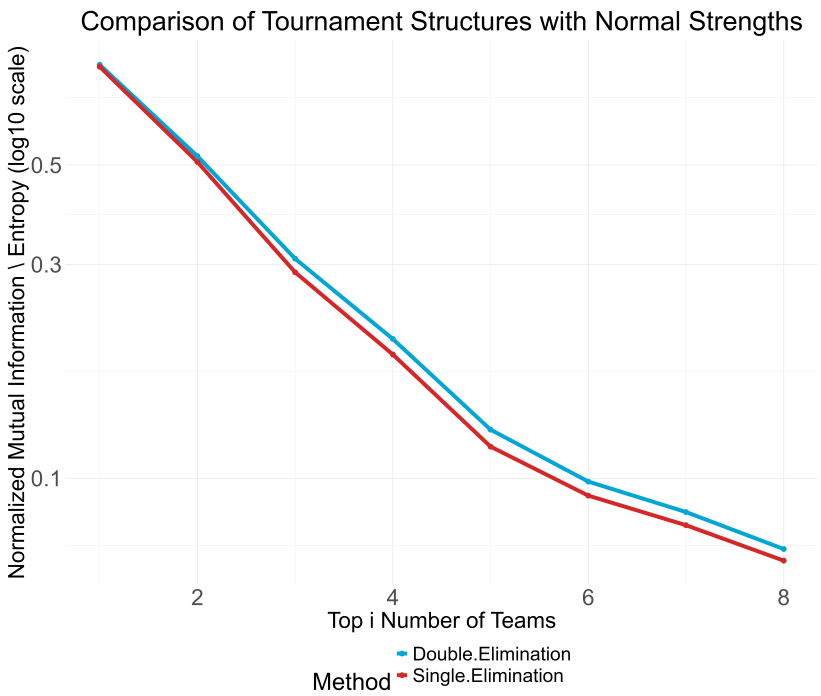

Figure 2: This figure focuses on the difference between single elimination and double elimination competition structures

respectively, in more detail as show in figure 1. Note that in terms of declaring a champion (i.e. when the number of salient ranks is 1), single and double elimination brackets are nearly identical is their ability to accomplish this. However, double elimination is slightly better than single elimination, with the gap growing wider as the number of salient ranks increases. One clear reason for this is that with double elimination the ranking of teams is "clearer". That is, in a single elimination tournament the two teams that lose in the semi-finals can equally make a claim as the third best team and in our schema they are randomly assigned to a rank fo 3 or 4. However, in a double elimination tournament, teams are definitively ranked 3 and 4 leading to beter outcomes when the number of salient ranks is in the range of 3 to 4 (or more).

4 Conclusion

This study introduced a weighted mutual information framework to evaluate the effectiveness of different competition formats in preserving the true underlying rankings of teams. We also introduce the concept of salient ranks, thereby extending and generalizing the concepts of efficacy and effectivity. By simulating outcomes of competition strucutres with varying seeding methods and applying an information-theoretic approach, we quantified how accurately each format conveys ranking information. Our findings confirm the intuitive advantage of round

robin tournaments, which consistently outperformed single and double elimination formats in ranking accuracy, but also quantify the difference between these structures. Additionally, we demonstrated the degree to which poor seeding structures can significantly degrade performance of a competition structure, in some cases making competition outcomes even worse than simply randomly generating a set of ranks.

These results underscore the trade-off between tournament accuracy and practical constraints such as time, cost, and entertainment value. While round robin offers the highest fidelity to true rankings, it is often impractical for large tournaments, highlighting the need for hybrid or adaptive structures that balance accuracy with logistical feasibility.

Future work should extend this analysis to scenarios with more tournament structures, varying numbers of competitors, incorporate probabilistic strength models reflecting real-world uncertainty, and explore alternative weighting schemes for different competitive priorities (e.g., 4 salient ranks versus 8 salient ranking). Applying these methods to actual tournament data could further validate their usefulness for organizers aiming to design fair and informative competitions.

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Supplementary Material

All supplementary material available at https://github.com/gjm112/tournaments.

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